

HPC Software II

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1 Introduction

For a 32×32 grid the max absolute error between Gathered Solution and Analytical Solution = $3.038285e - 01$. This was not a very good convergence and points to there being some error in the code. This was probably due to a similar issue that is shown with the black spots on the figure of the heatmap in the section below. I was unable to identify and fix the exact bug that was causing this.

2 Poisson Heatmap plot

The figure below shows the heatmap plot of the 1D poisson implementation. There are some black lines down the top right handside of the plot that I could not figure out. I am not sure why they occur there and why some are different lengths,

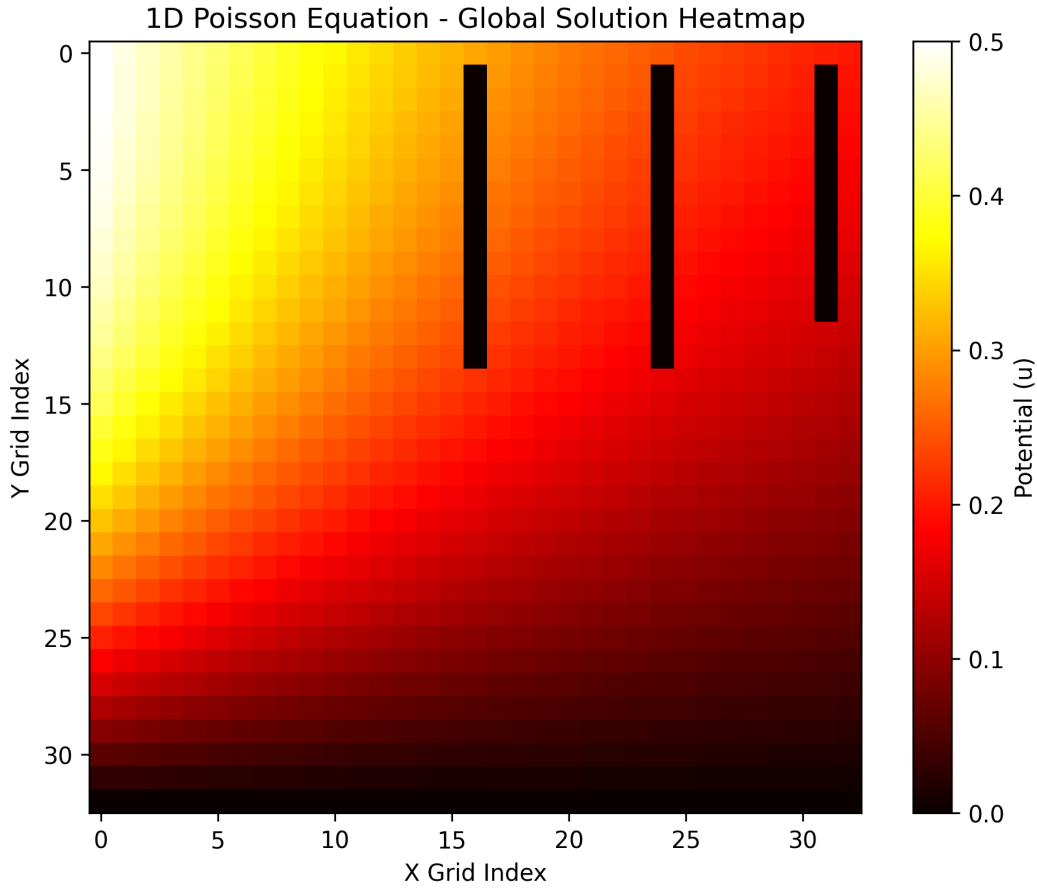


Figure 1: Heatmap plot of full grid after 1D poisson implementation

3 Grid size and Non-blocking comparisons

The table below shows that when the grid size was increased from 16x16 to 32x32 the absolute error decreased by an order of magnitude while the iterations increased by over 3.7x. This is pretty closely mirrored by the time increase of just under 3.5x. We can see when splitting the 32x32 grid over 16 processes instead of 4 the absolute error was the same. This is good as it shows the calculation done was the same but just splitting the grid over my processes. The time actually slightly increased but when it was spread over more processes. This might not hold over a larger grid size where the communication overhead would be a less significant portion of the overall computation time. On the small 32x32 grid the extra communication of the extra processes outweighs the benefit of spreading out the work.

Looking at the last two rows in the table we can see the only difference in the code was using Non-blocking sends and receives versus using traditional blocking send and receives. Again this gives the same absolute error which is a good sign that the calculation being performed was the same. There is a slight performance increase by using the non-blocking send and receives but it is very small at less than 0.0001 seconds difference. There is no

computation done in between the non-blocking sends and receive in the code therefore there is not any performance benefit.

Table 1: Performance and Validation Results for the 2D Parallel Poisson Solver

Grid Size	Procs	Topology	Exchange Method	Iterations	Time (s)	Max Abs. Error
16×16	4	2×2	SendRecv	249	0.003516	1.28×10^{-4}
32×32	4	2×2	SendRecv	929	0.012285	1.50×10^{-5}
32×32	16	4×4	SendRecv	929	0.012441	1.50×10^{-5}
32×32	16	4×4	Non-Blocking	929	0.012343	1.50×10^{-5}